

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and developed new user-facing features for a laboratory information system, enhancing functionality and user experience for environmental testing workflows.
- Automated data transfer processes between analytical instruments and the central LIMS, managing increasing data volumes and improving throughput by **30%**.
- Implemented data quarantine systems to ensure data integrity and compliance with quality control standards for laboratory results.
- Created and modified databases to support new functionalities within the laboratory information system, ensuring efficient data storage and retrieval.
- Optimized application performance for speed and scalability, reducing average response times by **20%** through code enhancements and query tuning.
- Collaborated with cross-functional teams in analytical, quality, and finance departments to understand data requirements and develop automated processes.

SKILLS

Languages – PHP, Python, SQL, Java, R

Web Technologies – REST APIs, HTML, JavaScript

Databases – PostgreSQL, MySQL, Oracle

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Development Tools – Git, Jenkins, Docker

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, RStudio

Methodologies – Agile, Scrum, Change Management

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Laboratory Information Management System (LIMS) Enhancement

- Developed new backend web applications and APIs to integrate instrument data streams with the core LIMS, supporting higher data throughput.
- Designed and implemented automated quality control verification routines within the LIMS, reducing manual checks by **40%** and improving accuracy.

- Refactored existing database schemas and optimized queries in PostgreSQL to support enhanced reporting and data computations for method validations.

Data Pipeline and Reporting Automation

- Built Python scripts to automate ETL processes for data import and export between lab instruments and the central data repository, improving data consistency.
- Created interactive dashboards in **Power BI** to visualize key performance indicators for laboratory operations, including bias, precision, and uncertainty of measurement.
- Documented release deployments and change management procedures for system updates, ensuring clear communication and adherence to standards.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)